

The CHEASE toroidal MHD equilibrium code: A brief user's guide

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Abstract

The Document aims to serve as a brief introduction for a new user of the CHEASE toroidal plasma equilibrium code as introduced in References [1, 2]. Details of the specification of equilibria and the required inputs to CHEASE are introduced and discussed; the equations solved and numerical methods used are discussed in References [1] and [2] and no attempt is made to cover these topics herein.

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1 Introduction

The Cubic Hermite Element Axisymmetric Static Equilibrium (CHEASE) code solves the elliptic second-order, nonlinear partial differential Grad-Shafranov equation¹ [5],

$$\nabla \cdot \frac{1}{R^2} \nabla \Psi = \frac{j_\varphi}{R} = -p'(\Psi) - \frac{1}{R^2} f f'(\Psi), \quad (1)$$

for a prescribed plasma boundary and prescribed radial pressure and current profiles. The above equation describes static, axisymmetric ideal MHD plasma equilibria. Here j_φ denotes the toroidal plasma current density, R the major radius of the torus and primes denote a derivative with respect to Ψ , the flux surface label. Throughout, it is assumed that the known plasma boundary is described by $\Psi = 0$ with $\Psi < 0$ everywhere within the plasma. The nature of the equilibria is determined by the two free functions $p'(\Psi)$ and $f f'(\Psi)$, which, along with the plasma boundary, must be configured in the call to CHEASE. The shape of the boundary is arbitrary and its specification is discussed in Section 3.1. The specification of the free functions are discussed in Sections 3.2 and 3.3.

Axisymmetric magnetic fields may be written in the familiar form

$$\mathbf{B} = f \nabla \varphi + \nabla \Psi \times \nabla \varphi. \quad (2)$$

Here Ψ is the poloidal magnetic flux, as defined by Equation 1 and φ is the geometric (ignorable) toroidal angle.

The CHEASE code has been developed and is maintained at CRPP, Lausanne. It is publicly available through a SVN server. A copy of the latest version can be obtained using

svn checkout <https://crppsvn.epfl.ch/repos/CHEASE/chease/trunk> chease

The version of the CHEASE code as described in Ref. [2] is available from the CPC Program Library [3].

2 Inputs to and initialisation of CHEASE

A CHEASE run is configured by defining and setting a number of parameters in either an XML file or a custom namelist file, the format of each is as discussed below. The current version of CHEASE first searches for the XML configuration file, which it expects to reside in the current working directory and to be called ‘chease_input.xml’. If this is not found, the current version of CHEASE then searches for a namelist configuration file called ‘chease_namelist’ which again should reside in the current working directory. The version of CHEASE available from the CPC Program library [3] supports only the namelist-style configuration, which it reads from standard input.

2.1 XML input

As part of the ITM (Integrated Tokamak Modelling) program the input schema to CHEASE has been changed to allow users to specify inputs to the code using XML files. These files are not discussed here: the namelist file discussed below is still valid.

2.2 Namelist input

The format of the CHEASE namelist configuration file is given in the Listing below. Note that this namelist is not the Fortran 90 standard namelist.

Listing 1: Structure of the input namelist read by CHEASE.

```
****
**** 4 character lines of maximum length 80
```

¹In fact, CHEASE solves the variational form of Eq. 1, cf. Eq. 26 of [2].

```

****
****
&EQDATA
<List of Namelist variables for CHEASE>
/
&NEWRUN
<List of Namelist variables for ERATO>
/
&END

```

The numerous variables which constitute the EQDATA namelist are introduced and discussed in the following Sections. Of primary interest in the present Document are the variables used to construct an equilibrium solution (Section 3) and those used to control the output formatting of this solution (Section 4). The NEWRUN namelist is used only for ERATO simulations and is not considered here. Thus, any namelist variable discussed in this Document may be assumed to be part of the EQDATA namelist, except if explicitly stated otherwise.

The list of namelist variables is a comma-separated list of key=value pairs. If a value to be passed is an array of values, the array is written as a comma separated list without any surrounding parentheses. The four character lines at the head of the file are used to label the run and must exist, even if they remain blank, since they will appear at the beginning of a number of output files. Comments may also appear within and between the lists of namelist variables and are preceded with an exclamation mark. An example namelist file is given in Section 3.5 and a list of all namelist variables, including a description, their default value and type is given in Appendix B.

The required equilibrium quantities, the plasma boundary and radial profiles of pressure and current, may be specified using analytical forms or, more commonly, using equilibria reconstructed from experimental data. In the former case, these quantities are specified through variables passed in the EQDATA namelist (cf. Section 3). In the later case, these equilibrium quantities are read in from an additional file, as either EXPEQ or EQDSK formatted files (see Appendix C), which, like ‘chease_namelist’, must reside in the current working directory and in both cases must be called EXPEQ.

3 Constructing an equilibrium

As previously stated, CHEASE computes ideal MHD equilibria for a prescribed plasma boundary and prescribed pressure and current profiles. These quantities must therefore be specified when calling CHEASE, as described in this Section. All namelist variables introduced in this Section are part of the EQDATA namelist except if explicitly stated otherwise.

3.1 Specification of the plasma boundary

The plasma boundary is assumed to be known by CHEASE and several methods exist for the specification of the boundary. The method used is controlled by the namelist parameter NSURF in the EQDATA namelist. The allowed values for NSURF are as follows.

Analytic specification of the plasma boundary

NSURF=1: analytic Solov’ev equilibrium [6].

This class of equilibria are characterized by the following specification of the two required free functions:

$$\begin{aligned}
 p(\Psi) &= -\frac{1 + \kappa^2}{\kappa R_0^3 q_0} \Psi \\
 f(\Psi) &= 1
 \end{aligned}$$

i.e. by a right-hand side of the Grad-Shafranov equation, Eq. 1, that is linear in Ψ . Here Ψ is given by

$$\Psi = \frac{\kappa}{2R_0^3q_0} \left[\frac{R^2Z^2}{\kappa^2} + \frac{1}{2} (R^2 - R_0^2)^2 - a^2R_0^2 \right] \quad (3)$$

and the plasma boundary is defined by

$$R(\theta) = R_0\sqrt{1 + 2\epsilon\cos\theta}, \quad (4)$$

$$Z(\theta) = \frac{R_0\epsilon\kappa\sin\theta}{\sqrt{1 + 2\epsilon\cos\theta}}. \quad (5)$$

The various quantities which define the plasma boundary and their corresponding namelist names (given in uppercase) are: ASPCT= $\epsilon=a/R_0$ is the inverse aspect ratio of the torus, RC= R_0 is the plasma major radius, ELONG= κ is the plasma elongation and CQ0= q_0 is the safety factor at the magnetic axis. Here a is the plasma minor radius.

NSURF=2 and NSURF=4: generalized INTOR formula

These two options use different forms of the generalised INTOR formula²

$$R = R_0 + R_0a\cos(\theta + \delta\sin\theta - \xi\sin 2\theta)(1 + b\cos\theta) \quad (6a)$$

$$Z = R_0a\kappa g(\theta)\sin(\theta + \zeta\sin 2\theta) \quad (6b)$$

where

$$g(\theta) = 1 + \frac{\sigma}{1 + \{\sin^2[1/2(\theta - \theta_0)] + \bar{\xi}\}^\nu / \Delta} \quad (7)$$

is included to allow one to model diverted plasma boundaries [7]. Here the minor radius a is determined by demanding that the user-specified inverse aspect ratio, ASPCT, is given by

$$\epsilon = \frac{a}{R_0} = \frac{R_{\max} - R_{\min}}{R_{\max} + R_{\min}}, \quad (8)$$

with $R_{\min}$ and $R_{\max}$ respectively the minimum and maximum radial extents of the plasma boundary.

NSURF=2: Up-down symmetric equilibria ("Racetrack" equilibria)

$$R = R_0 + R_0a\cos(\theta + \delta\sin\theta - \xi\sin 2\theta)(1 + b\cos\theta) \quad (9a)$$

$$Z = R_0a\kappa\sin(\theta + \zeta\sin 2\theta) \quad (9b)$$

In this case $g(\theta) = 1$ and the equilibrium is specified by the namelist variables: RC= R_0 , TRIANG= δ is the triangularity, ELONG= κ , ASPCT= $\epsilon=a/R_0$, BEANS= b [is what], XI= ξ [is what], CETA= ζ [is what].

NSURF=4: Up-down asymmetric equilibria (SND equilibria)

$$R = R_0 + R_0a\cos(\theta + \delta\sin\theta)(1 + b\cos\theta) \quad (10a)$$

$$Z = R_0a\kappa g(\theta)\sin(\theta) \quad (10b)$$

In this case $\xi = \zeta = 0$ and the equilibrium is specified by the namelist variables: RC= R_0 , TRIANG= δ , ELONG= κ , ASPCT= $\epsilon=a/R_0$, BEANS= b , THETA0= θ_0 (the poloidal angle of the X-point in radians), RNU= ν is the exponent in the denominator of $g(\theta)$, XI= $\bar{\xi}$ (X-point 'sharpness'), SGMA= σ [is what] and DELTA= Δ [is what].

²Note that Eq. 6 corrects the expression given in Ref. [2].

Figure 1: Examples of analytic equilibria which may be used in CHEASE and configured through the CHEASE namelist. Left: ‘Racetrack’ equilibria. Specified with NSURF=2 this plasma boundary is up-down symmetric. This particular example uses Right: Single-null diverted (SND) plasma boundary. Specified with NSURF=4, this analytic plasma boundary allows one to specify up-down asymmetric equilibria which includes a single null-point. This particular example uses . . .

Empirical plasma boundary

NSURF=6: Numerical plasma boundary

Alternatively, the plasma boundary may be determined empirically and therefore may not have an analytic representation. In this case the boundary may be specified numerically as a list of pairs of {R,Z} coordinates which define the boundary.

If NEQDSK = 0: the boundary information is read from the file EXPEQ, whose format is described in Appendix C.1 of this Document:

If NEQDSK = 1: the data are read in from EQDSK file, the standard input/output file format of EFIT [8], as described in Section 5.4.6 of Ref. [2] and in Appendix C.2 of this Document.

3.2 Specification of the pressure profile

The required pressure gradient profile may be specified analytically or numerically.

Analytic specification of the pressure gradient profile

Each of the analytical forms of the pressure gradient profile are prescribed as functions of the normalised flux: $s = s(\Psi) = \sqrt{1 - \Psi/\Psi_{\min}}$. The choice of analytic form is controlled by the NPPFUN namelist variable:

NPPFUN=1: polynomial form of p'

In this case p' is specified as an polynomial of the form

$$p'(\Psi) = \sum_i^{\text{NSOUR}} \text{AP}(i) \left[1 - \frac{\Psi}{\Psi_{\min}} \right]^{(i-1)} \quad (11)$$

with NSOUR and AP namelist variables with NSOUR and **size**(AP) ≤ 10 .

NPPFUN=2: p' specified as a combination of polynomials

In this case p' specified as a combination of polynomials over three different sub intervals of the normalised flux label. This case seems a bit arcane to me, so I have not tried to describe the namelist inputs for this case. The reader is directed to Section 6.4.2 of Ref. [2] for more information on this particular case.

NPPFUN=3: [what to call this case]

In this case p' is specified as

$$p'(\Psi) = p_0 \left[1 - \left(1 - \frac{\Psi}{\Psi_{\min}} \right)^\beta \right]^\alpha \quad (12)$$

with p_0 , α and β specified through the namelist variable AP as AP(1)= p_0 , AP(2)= α and AP(3)= β .

Empirical pressure gradient profile

NPPFUN = 4: p' read from file

In this case the pressure gradient profile is specified numerically and the namelist variable NEQDSK again controls the file format (see plasma boundary specification above and Appendices C.1 & C.2).

3.3 Specification of the current profile

The toroidal current density j_φ may be defined in terms of a number of other quantities, as described in Section 2.2 of Ref. [2]. These allowed definitions are in terms of ff' , one of the free functions appearing in the Grad Shafranov equation, I^* , the surface averaged current density, or $I_\parallel$, the averaged parallel current density. This choice of definition is controlled by the NSTTP namelist variable: NSTTP=1 indicates that j_φ is to be defined in terms of ff' ; NSTTP=2 indicates that j_φ is to be defined in terms of I^* ; NSTTP=3 indicates that j_φ is to be defined in terms of $I_\parallel$. In the following, we let $\mathcal{I}$ stand for ff' , I^* or $I_\parallel$ depending on our choice of NSTTP.

The functional dependence of j_φ is specified by the NFUNC namelist variable.

Analytic specification of the current density profile

As with the definitions of the pressure gradient profile, the current density profile is specified as a function of the normalised radial coordinate $s(\Psi) = \sqrt{1 - \Psi/\Psi_{\min}}$.

NFUNC = 1: polynomial form of $\mathcal{I}$

In this case $\mathcal{I}$ is specified as an polynomial in the same way that p' was specified in the first of the pressure gradient cases above.

$$\mathcal{I}(\Psi) = \sum_{i=1}^{\text{NSOUR}} \text{AT}(i) \left[1 - \frac{\Psi}{\Psi_{\min}} \right]^{(i-1)} \quad (13)$$

with NSOUR and AT namelist variables with NSOUR and **size**(AT) ≤ 10 . Note that the order of the polynomial, NSOUR, is the same as for the pressure gradient profile, whilst the coefficients (AP for the pressure gradient, AT for the current density) may differ.

NFUNC=2: $\mathcal{I}$ specified as a combination of polynomials

As above, this choice of the parameter NFUNC is rather arcane to me and the reader is once again directed to Section 6.4.2 of Ref. [2] for more information on this particular case.

NFUNC=3: [what to call this case]

In this case $\mathcal{I}$ has the same functional form as the NPPFUN=3 case for the pressure gradient:

$$\mathcal{I}(\Psi) = \mathcal{I}_0 \left[1 - \left(1 - \frac{\Psi}{\Psi_{\min}} \right)^\delta \right]^\gamma \quad (14)$$

with $\mathcal{I}_0$, γ and δ specified through the namelist variable AT as AT(1)= $\mathcal{I}_0$, AT(2)= γ and AT(3)= δ .

Empirical current density profile

NFUNC = 4: $\mathcal{I}$ read from file

In this case the $\mathcal{I} = \mathcal{I}(\Psi)$ profile is specified numerically and the namelist variable NEQDSK again controls the file format (see pressure gradient specification above and Appendices C.1 & C.2).

3.4 Specification of the Bootstrap current fraction

In addition to the prescribed pressure gradient and current profiles, one may compute an equilibrium solution with a specific fraction of bootstrap current, $\langle \mathbf{J} \cdot \mathbf{B} \rangle_{\text{bs}} / \langle \mathbf{J} \cdot \mathbf{B} \rangle$, which is either prescribed as a constant or as a function of the flux label, $s = \sqrt{1 - \Psi/\Psi_{\text{min}}}$, in much the same way as the current and pressure profiles previously. In order to configure CHEASE to run with a specified bootstrap current (SBC) fraction the namelist variable NBSOPT must be 1 (it is 0 by default). The functional form of $\mathbf{j}_{\text{bs}}$ is then specified by the namelist parameter NBSTRP.

NBSTRP = 1: Constant SBC fraction

In this case the SBC fraction is constant, with a value specified by the namelist parameter BSFRAC.

NBSTRP = 2: Radial variation of the SBC fraction

In this case, the radial profile of SBC fraction is specified in terms of a flux function $C(s)$. This function and its functional dependence on s is specified in the same manner as the pressure gradient above but with ... [2] - *the CPC manual is not clear enough here*

3.5 Example namelist file

Based on the information above, we may now construct, as an example, a minimal input namelist to CHEASE which uses analytic prescriptions for the required physical quantities. We attempt to construct a MAST-like spherical tokamak Solov'ev-type equilibrium. Here we do not specify the bootstrap current fraction.

Listing 2: Minimal input namelist to CHEASE which describes an analytical Solov'ev plasma boundary and linear equilibrium (pressure gradient and current) profiles.

```
**** Minimal input namelist to CHEASE which describes an analytical MAST-like
**** equilibrium with an inverse aspect ratio of ?, a major radius of ?
**** and an elongation of ?, with linear pressure gradient and current
**** density profiles.
```

```
&EQDATA
```

```
! Quantities related to the plasma boundary:
```

```
NSURF = 1,
```

```
RC = 0.964,
```

```
ASPCT = 0.66667, ! This is an example of a comment which is
```

```
ELONG = 1.85, ! within the list of namelist variables.
```

```
CQ0 = 1.0,
```

```
NSOUR = 2,
```

```
AP = 0.0,1.0, ! These lines demonstrate how to specify an
```

```
AT = 0.0,1.0, ! array or list of values.
```

```
! Quantities related to pressure gradient profile:
```

```
! Quantities related to current profile:
```

```
! Quantities related to output file formatting.
```

```
! See Section 4.3 for more information.
```

```
/
```

```
&NEWRUN
```

```
! ERATO namelist variables: this section may be left blank or
```

```
! omitted all together.
```

```
/
```

```
&END
```

4 Configuring output from CHEASE

CHEASE, by default, outputs a number of different files of various formats. The particular formats and coordinate systems used in these files may be configured during the call to CHEASE with the NIDEAL namelist variable. Each of the allowed values of this option uses a different mapping of the equilibrium solution as computed by CHEASE. Typically each value corresponds to a set of output files as required by a different code with its own internal coordinate system. The following list extends the documented values of NIDEAL in Ref. [2] to those accepted by the present version of CHEASE. This list is likely to be extended in the future. However, the existing values should not change. Unique files generated for each of the NIDEAL options are indicated.

NIDEAL= 0: MARS

NIDEAL= 1: ERATO

NIDEAL= 2: LION

NIDEAL= 3: NOWA-W and PEST (file INP1)

NIDEAL= 4: PENN (file NPENN)

NIDEAL= 5: XTOR (Astro file OUTXTOR)

NIDEAL= 6: Default case

Standard file outputs including the files EQDATA.OUT and EQDSK.OUT

NIDEAL= 7: For various gyrokinetic codes with straight field Jacobians, e.g. ORB5 and GENE (file EQCIN)

NIDEAL= 8: ELITE (file NELITE)

NIDEAL= 9: OGYROPSI (file OGYROPSI)

Requires straight field Jacobian [not sure about this]

NIDEAL=10: For GKW and NEOART with Hamada coordinates

Files outputted include ‘neoart.dat’ and ‘hamada.dat’.

4.1 Output file formats

The output file formats are not particularly well documented in Ref. [2] and it is beyond the scope of this Document to attempt to introduce each format. The files EQDATA.OUT and EQDSK.OUT are EQDATA and EQDSK files respectively and are therefore documented in Appendix C. Here we simply discuss the contents of the files ‘neoart.dat’, which contains the radial profiles of a number of flux surface averaged quantities, and ‘’, which contains the poloidal projection of the equilibrium flux surfaces.

4.1.1 neoart.dat

The output file ‘neoart.dat’, generated when NIDEAL=10 in the input namelist, contains a number of flux surface averaged quantities, which may be passed to/and are required by transport codes such as NEOART [9] and NCLASS [10].

4.1.2 unknown.dat

The most accessible output file generated by CHEASE is the relatively self-documenting HDF5 file: ?.dat. This file is generated when NIDEAL=? and provided that CHEASE has been compiled with the HDF5 library and FUTILS. Note that this file is not written with the original CPC library version of CHEASE.

4.1.3 ogyropsi.dat

[Describe file format...]

...are outputted on a rectilinear two-dimensional grid with coordinates (ψ, χ) , where $\psi = 2\pi\Psi_p$ and

$$\chi(\theta) = \frac{1}{q} \int_0^\theta \frac{\mathbf{B} \cdot \nabla \zeta}{\mathbf{B} \cdot \nabla \theta'} d\theta', \quad (15)$$

with θ the geometric poloidal angle.

4.2 Scaling of the equilibrium

It is well known that a single solution of the Grad-Shafranov Equation (1) describes a family of equilibria, with fixed poloidal beta but different plasma current and rotational transform values [2]. CHEASE, while solving Eq. 1, assumes $f(\Psi = 0) = 1$. Thus, it is important when using experimental or numerical data to configure CHEASE that this data was also obtained or constructed with this assumption.

If this was not the case, the solution generated by CHEASE may be scaled to match the actual value of f at some specified flux surface. Two transformations which leave the Grad-Shafranov equation unchanged are

1. $\Psi_{\text{new}} = h\Psi$, $f_{\text{new}} = hf$, $p_{\text{new}} = h^2p$, and
2. $f_{\text{new}}^2 = T^2 + k$,

with h and k two constants. The second of these transformations leaves Ψ and p unchanged. CHEASE uses these transformations to generate an equilibrium solution with prescribed values of either the total current

$$I = \oint_{\Psi=0} j_\varphi dS_p \quad (16)$$

or the safety factor,

$$q(\Psi_q) = \frac{f}{2\pi} \oint \frac{dl}{r|\nabla\Psi|}, \quad (17)$$

specified on a given flux surface Ψ_q . These transformations also allow the specification of f on a given flux surface Ψ_f .

The simplest available transformation in CHEASE is to specify that f is one either on the magnetic axis or at the plasma boundary. This is controlled by the namelist variable NTMF0:

NTMF0 = 0 (default value):

In this case $f(\Psi = 0) = 1$ is assumed.

NTMF0 = 1:

In this case $f(\Psi = \Psi_{\text{min}}) = 1$ is assumed.

This transformation is made in conjunction with one of the following transformations, which require that we specify some of the namelist variables NCSCAL, CURRT, CSSPEC and/or QSPEC. The type of scaling is specified by NCSCAL:

NCSCAL=2 (default value): Specification of total current

In this case, the equilibrium solution is scaled such that the total current is equal to the value of the namelist variable CURRT. Note that CURRT is not specified in units of Amperes but as $\mu_0 I$. For example, specifying a total plasma current of 0.5 MA, one would set $\text{CURRT} = 5 \times 10^5 \times \mu_0 \simeq 0.6283$. *The value of CURRT is then normalised:* from Appendix A we therefore set

$$\text{CURRT} = \frac{\mu_0 I_P}{R_0 B_0}. \quad (18)$$

This transformation is applied as follows: first, transformation 1 (above) is applied, with $h = I_{\text{specified}}/I = \text{CURRT}/I$ and then ...

NCSCAL=1 or 3: Specification of safety factor

In this case, the equilibrium is scaled such that the safety factor, Eq. 17, matches the value QSPEC, specified at the radial location CSSPEC, i.e. $q(\text{CSSPEC}) = \text{QSPEC}$. [Describe the scaling: pp. 231 of [2]] The definition of the radial variable differs depending on the value of NCSCAL:

If NCSCAL= 1:

In this case, the radial coordinate is simply

$$s(\Psi) = \sqrt{\frac{|\Psi_{\text{edge}} - \Psi|}{|\Psi_{\text{edge}} - \Psi_0|}} \quad (19)$$

This does not seem to be the case! The above formula would suggest that $s \in [1, 0]$ with the magnetic axis defined by $s = 1$. However, from running CHEASE it seems that $s \in [0, 1]$ with $s = 0$ defining the magnetic axis.

To avoid confusion when specifying the safety factor normalisation use NCSCAL = 3 (see below).

If NCSCAL = 3:

In this case, the radial coordinate and CCSPEC corresponds to the generalised radial coordinate

$$\rho(\Psi) = \sqrt{\frac{V(\Psi)}{V_{\text{total}}}} \quad (20)$$

Here $\rho \in [0, 1]$ with the magnetic axis defined by $\rho = 0$ and the LCFS defined by $\rho = 1$.

[Note: [2] is not clear enough here. It seems to suggest that the user can specify the value of f at the same location as the specified safety factor. However, is this specified value of f just the value required for consistency with the scaling of other parameters?]

4.2.1 The output files NIN and NOUT

The files NIN and NOUT, written during each run of CHEASE regardless of the value of the NIDEAL namelist variable, allow the user of CHEASE to take advantage of the equilibrium scaling laws described in the current Section.

From [2]: Every quantity required for the reconstruction of the converged solution of Eq. (26) is stored into the file NOUT before the rescaling of the equilibrium. This file can be re-utilized for a subsequent equilibrium calculation which only differs from the first by another scaling (see above). If NOPT $\neq 0$ in the Namelist, CHEASE reads an equilibrium from the file NIN, which is organized in an identical way to NOUT.

5 Additional topics

There are a number of other topics which still need to be covered. Topics including packing and scaling of the equilibrium solution. These are discussions I hope to include here in the future. For now the reader is directed to Sections 2.3, 3, 5.3 and 5.4 of Ref. [2].

5.1 Coordinate transformations

5.2 Mesh packing

A CHEASE normalisations

The following was adapted from a document written by O. Sauter on 8 May, 2009.

From Equation 2 we have that

$$[B] = \frac{[f]}{[R]} \rightarrow f_{\text{CHEASE}} = \frac{f_{\text{phys}}}{B_0 R_0} \quad (21)$$

Where R_0 and B_0 are the values used for normalizing the major radius and magnetic field respectively:

$$R_{\text{CHEASE}} = \frac{R_{\text{phys}}}{R_0}, \quad B_{\text{CHEASE}} = \frac{B_{\text{phys}}}{B_0}. \quad (22)$$

If one uses the boundary conditions $f_{\text{CHEASE}}(\Psi = 0) = 1$, then B_0 is the vacuum magnetic field at $R = R_0$ such that $f(\Psi = 0) = R_0 B_0$. Furthermore, we have from Equation 2 that the units of the poloidal flux Ψ are

$$[\Psi] = [B][R]^2. \quad (23)$$

Therefore, we have that

$$\Psi_{\text{CHEASE}} = \frac{\Psi_{\text{phys}}}{B_0 R_0^2}. \quad (24)$$

Current

From

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} \quad (25)$$

we have that the units of current density are

$$[j] = \frac{[B]}{\mu_0 [R]}, \quad (26)$$

which gives the following normalization for the current density

$$j_{\text{CHEASE}} = j_{\text{phys}} \frac{\mu_0 R_0}{B_0}, \quad (27)$$

and the corresponding relation for the total current

$$[I] = \frac{[B][R]}{\mu_0} \rightarrow I_{\text{CHEASE}} = I_{\text{phys}} \frac{\mu_0}{R_0 B_0}. \quad (28)$$

The units of $f f'$ are

$$[f f'] = \left[f \frac{\partial f}{\partial \Psi} \right] = \frac{[f]^2}{[\Psi]} = B_0, \quad (29)$$

which yields the normalisation

$$f f'_{\text{CHEASE}} = \frac{f f'_{\text{phys}}}{B_0}. \quad (30)$$

Pressure

From

$$\nabla p = \mathbf{j} \times \mathbf{B} \quad (31)$$

we have that the units of pressure are

$$[p] = [j][B][R] = \frac{B_0}{\mu_0 R_0} B_0 R_0 = \frac{B_0^2}{\mu_0}. \quad (32)$$

Then we define

$$p_{\text{CHEASE}} = p_{\text{phys}} \frac{\mu_0}{B_0^2}. \quad (33)$$

The units of pressure gradient are similarly found to be

$$[p'] = \left[\frac{\partial p}{\partial \Psi} \right] = \frac{[p]}{[\Psi]} = \frac{B_0}{\mu_0 R_0^2}, \quad (34)$$

which yields the normalisation

$$p'_{\text{CHEASE}} = p'_{\text{phys}} \frac{\mu_0 R_0^2}{B_0}. \quad (35)$$

[and the rest ...]

B EQDATA namelist variables

In this Section we summarise the EQDATA namelist variables and their default values.

Table 1: A summary of all EQDATA namelist variables, including a description of each flag, its type and default value.

Variable	Type	Default	Description
Equilibrium specification variables			
NSURF	int	1	Select method for specifying plasma boundary (cf. Section 3.1)
NPPFUN	int	1	Select functional form of $p'(s)$ (cf. Section 3.2)
NSTTP	int	1	Select functional form of $\mathcal{I}(s)$ (cf. Section 3.3)
NSOUR	int	2	If NFUNC=1 or NPPFUN=1, this is the degree of the polynomial used to define the con
Output configuration variables			
Optimisation and mapping variables			

Run control parameters

epsilon relax ninmap ninsca nmgaus nskaus ntgaus

Mesh parameters

aplace awidth bplace bwidth cplace cwidth dplace dwidth eplace ewidth qplace qwidth solpda solpdb solpdc
solpdd solpde msmax nchi ndift negp ner niso nmesha nmeshb nmeshc nmeshd nmeshe npoida npoidb npoidc
npoidd npoide npoidq npsi ns nt ntnova nv nvexp

Equilibrium control parameters

bsfrac cfbal cfiness cfinesso cpress cpresso cq0 csspec currt etaei gamma pangle predge psisl qspec rzion
scalne scexp nbsfun nbsopt nbstrp nfunrho nrfp

Plasma boundary parameters

aspet beans ceta delta elong rc rnu rz0 r0 sigma theta0 triang triplt xi nsurf nsym

Profiles parameters

afbs afbs2 ap ap2 at at2 at3 at4 nfunc npr npp nppfun nppr nprofz npropt nsour nsttp

input/output control parameters

bentaxis bentqprofile bentradius b0exp qvalneo rboxlen rboxlft rext rpeop rz0w r0exp r0w signb0xp signipxp
slimit snumber spitzer treeitm zboxlen nanal nbal nblc0 nblopt nbpsout nbsexpq ncscal ndiagop ndifps
neonbqs neqdsq neqdxtpo nfftpt nideal nitmopt nitmrun nitmshot nopt nplot nprpsi nrbox nrscal nsmooth
ntcase ntest ntmf0 nturn nzbox

C Experimental data file formats

If CHEASE is configured to read the plasma boundary and radial profile information from file (with the specification of NSURF=6, NPPFUN=4 and/or NFUNC=4 in the EQDATA namelist in the chease namelist file, cf. Sections 3.1–3.3) then either of the EXPEQ or EQDSK file formats may be used to specify this information. This option is typically used when CHEASE is used to solve the Grad Shafranov equation with empirically determined profiles and plasma boundary.

In this case an additional file must be located in the current working directory along with the chease namelist or XML file. This additional file is either a EXPEQ or EQDSK file although, confusingly, this file must be named EXPEQ in either case. The format of the data contained within the file is passed to CHEASE as the NEQDSK namelist variable, with NEQDSK=0 indicating that the file EXPEQ is in EXPEQ format and NEQDSK=1 indicating that the file EXPEQ is in EQDSK format. Each of these file formats are now discussed in turn.

C.1 EXPEQ file format

The EXPEQ file format is the simpler of the two allowed input file formats and contains the radial profiles of the pressure gradient and current function and the {R,Z} locations of the plasma boundary. It is important to note that *all* quantities are specified in normalised CHEASE units (see Appendix A).

Listing 3: EXPEQ file format specification.

```
inverse_aspect_ratio
Z_0
edge_pressure
nbound
Rbound(1)      Zbound(1)
Rbound(2)      Zbound(2)
Rbound(3)      Zbound(3)
...
Rbound(nbound) Zbound(nbound)
NPPF1
NPROPT NRHOMESH
s(1)
s(2)
s(3)
...
s(NPPF1)
RPPF(1)
RPPF(2)
RPPF(3)
...
RPPF(NPPF1)
RFUN(1)
RFUN(2)
RFUN(3)
...
RFUN(NPPF1)
```

The exact definitions of the above quantities are as follows:

Specification of scalar quantities

inverse_aspect_ratio: The inverse aspect ratio of the last closed flux surface, defined as

$$\epsilon = \frac{R_{\max} - R_{\min}}{R_{\max} + R_{\min}}, \quad (36)$$

with $R_{\min}$ and $R_{\max}$ the minimum and maximum major radial extents of the last closed flux surface in the midplane.

Z_0: The ratio of the vertical to major radial geometric axis position,

$$Z_0 = \frac{Z_{\max} + Z_{\min}}{R_{\max} + R_{\min}}, \quad (37)$$

with $Z_{\min}$ and $Z_{\max}$ the minimum and maximum vertical extents of the last closed flux surface.

edge_pressure: The normalised static edge plasma pressure, i.e. $p_{\text{edge}} = \mu_0 p(\Psi = 0)/B_0^2$.

Plasma boundary specification

nbound: Number of points used to describe the last closed flux surface.

Rbound and Zbound: A list, of length *nbound*, of pairs of major radial and vertical coordinates of the last closed flux surface.

Radial profile specifications

NPPF1: Number of points used in the specification of each of the radial profiles.

NPROPT: Specify the definition of the current profile (cf. the NSTTP namelist variable, discussed in Section 3.3). Note, the NSTTP namelist variable is ignored if the profile specification is read from an EXPEQ file.

NRHOMESH: Specifies the definition of the radial coordinate (see ψ below). If zero the radial coordinate is defined in terms of the poloidal flux, if one the radial coordinate is defined in terms of the toroidal flux. In the present version of CHEASE NRHOMESH can only be non-zero if $\text{NSTTP} \geq 2$, $\text{NEQDSK} = 0$ and $\text{NITMOPT} < 10$.

s: The normalised radial coordinate grid, on which the pressure gradient and current density profiles are specified. This coordinate is defined as

$$s = \sqrt{\frac{\psi - \psi_{\text{axis}}}{\psi_{\text{edge}} - \psi_{\text{axis}}}}, \quad (38)$$

with ψ either the poloidal or toroidal flux, depending on the value of NRHOMESH.

RPPF: Normalised pressure gradient profile, $p' = (\mu_0 R_0^2/B_0)\partial p/\partial\psi$, with the derivate taken with respect to ψ , not s .

RFUN: Current density specification, the exact specification of which depends on the value of NPROPT.

All variables are floating point numbers except nbound, NPPF1, NPROPT and NRHOMESH which are integers. Integers are written to file with a width of five characters and floats are written to file with a width of 18 character and with eight decimal places.

We note that the use of two integers NPROPT and NRHOMESH to specify the definition of the radial profiles (see above) differs from that defined in Reference [2], in which only one integer is discussed. The specification outlined in the present Document is the format expected by CHEASE: since Reference [2] was published the variable NRHOMESH has been added to the EXPEQ file.

Finally, we note that capitalised variable names above indicate that these variable names have been chosen to be the same as the corresponding CHEASE variable names.

C.2 EQDSK file format

The EQDSK format is a standard input/output file format used with, for example, EFIT [8]. However, a large number of variations of this file format exists each describing different aspects of the equilibrium solution. Each variation is usually distinguished by a single letter prefixing EQDSK. For example, there exists the A EXPEQ, E EXPEQ and G EQDSK file formats. It is the G EQDSK format which is read and outputted by CHEASE and this format is now described.

G EQDSK file format

G EQDSK files contains more information than the EXPEQ file format, including the full poloidal cross-section of the poloidal flux (not just the bounding flux surface) evaluated on a rectangular grid and various (additional) radial functions such as the plasma pressure and safety factor. Information on the plasma boundary and the surrounding limiter contour is also provided. Unlike the EXPEQ file format, **all quantities in the EQDSK file are in MKSA units.**

Listing 4: EQDSK file format specification.

```
' comments and date', i3, nr, nz
Rboxlen, Zboxlen, R0, Rmin, Z0
Raxis, Zaxis, Psi_axis, Psi_bound, B0
current, Psi_axis, xdum, Raxis, xdum
Zaxis, xdum, Psi_bound, xdum, xdum
f(1)
f(2)
...
f(nr)
pressure(1)
pressure(2)
...
pressure(nr)
ffprime(1)
ffprime(2)
...
ffprime(nr)
pprime(1)
pprime(2)
...
pprime(nr)
psi(1,1) psi(1,2) ... psi(1,nz)
psi(2,1) psi(2,2) ... psi(2,nz)
...
psi(nr,1) psi(nr,2) ... psi(nr,nz)
q(1)
q(2)
...
q(nr)
nbound nlimiter
Rbound(1) Zbound(1)
Rbound(2) Zbound(2)
...
Rbound(nlimiter) Zbound(nlimiter)
nbound
```



```

Rbound(1)      Zbound(1)
Rbound(2)      Zbound(2)
...
Rbound(nlimiter) Zbound(nlimiter)

```

The exact definitions of the above quantities are as follows:

Specification of scalar quantities (some quantities are repeated, not sure why)

nr: Number of major radial grid points

nz: Number of vertical grid points

Rboxlen: **Extent** of the poloidal cross section grid in the major radial direction [m]

Zboxlen: Extent of the poloidal cross section grid in the vertical direction [m]

Rmin: Minimum value of the major radius in the poloidal cross section grid [m]

R0: A measure of the ‘central’ major radial position [m] (CHEASE writes R0EXP to this variable)

Z0: Vertical position of the centre (**elevation**) of the poloidal cross section grid [m]

Raxis: Major radial position of the magnetic axis [m]

Zaxis: Vertical position of the magnetic axis [m]

B0: Vacuum toroidal field evaluated at R0 [T] (CHEASE writes B0EXP to this variable)

Psi_axis: Value of the **poloidal flux** on the magnetic axis [Tm^2]

Psi_bound: Value of the poloidal flux on the plasma boundary (**last closed flux surface**) [Tm^2]

current: Total plasma current [A]

xdum: Undefined variable. Simply set to zero by CHEASE

Radial profile specification: Each of these profiles is specified on a uniform, **equidistant** poloidal flux grid with nr-1 intervals and with the poloidal flux in the range [Psi_axis Psi_bound].

f: ‘Poloidal current’ profile, $f = RB_\phi$ [Tm]

pressure: Plasma pressure profile [Pa]

ffprime: $ff'(\Psi)$ [T]

pprime: Derivate of the plasma pressure with respect to the poloidal flux coordinate [$\text{PaT}^{-1}\text{m}^{-2}$]

q: Tokamak safety factor [-]

Poloidal flux specification

psi: The **poloidal flux** [Tm^2], defined on a uniform rectangular grid of major radius and vertical position values. The grid on which the poloidal flux is defined is specified by the quantities nr, nz, Rboxlen, Zboxlen, Rmin and Z0. The coordinates of the grid points are constructed as:

$$\begin{aligned}
 R &= R_{\min} + \frac{R_{\text{boxlen}}}{nr - 1} \times i, \quad i = 1 : nr \\
 Z &= Z_0 - \frac{Z_{\text{boxlen}}}{2} + \frac{Z_{\text{boxlen}}}{nz - 1} \times j, \quad j = 1 : nz
 \end{aligned}$$

Plasma boundary specification

nbound: number of pairs of coordinates used to describe the plasma boundary (last closed flux surface)

Rbound and Zbound: A list, of length nbound, of pairs of major radial and vertical coordinates of the last closed flux surface [m].

Limiter specification

nlimiter: number of pairs of coordinates used to describe the limiter

Rbound and Zbound: A list, of length nlimiter, of pairs of major radial and vertical coordinates of the limiter. In CHEASE, this limiter boundary is simply set to the boundary of the rectangular grid, defined above [m].

References

- [1] Lütjens H, Bondeson A and Roy A, . . . , Computer Physics Communications **69** 287 (1992)
- [2] Lütjens H, Bondeson A and Sauter O, *The CHEASE code for toroidal MHD equilibria*, Computer Physics Communications **97** 219–260 (1996)
- [3] CPC Program Library, Queen’s University of Belfast, N. Ireland
Catalogue identifier: ADDH
- [4] CHEASE public svn: <https://crppsvn.epfl.ch/repos/CHEASE/chease/trunk>
- [5] Shafranov V D, ZhETF **33** 710 (1957); Sov. Phys. JETP **8** 494 (1958); Lüst R and A. Schlüter A, Z. Naturforsch **129** 850 (1957); Grad H and Rubin H, Proc. 2nd Int. Conf. on the Peaceful Uses of Atomic Energy, Vol. 31 (United Nations, Geneva, 1958) p. 190.
- [6] Solovév L S, Zh. Tekh. Fiz. **53** 626 (1967); JETP **26** 400 (1968)
- [7] Roy A and Troyon E, Theory of Fusion Plasmas, Proc. Joint Varenna-Lausanne Int. Workshop, 1988 (Editrice Compositori, Bologna, 1988) p. 175.
- [8] Lao L U et al., Nucl. Fusion **30** 1035 (1990)
- [9] Peeters A G, *Reduced charge state equations that describe Pfirsch Schluter impurity transport in tokamak plasma*, Phys. Plasmas **7** pp. 268–75 (2000); Dux R, Peeters A G, Gude A, Kallenbach A, Neu R, ASDEX Upgrade Team, *Z dependence of the core impurity transport in ASDEX Upgrade H mode discharges*, Nuclear Fusion **39** pp. 1509–22 (1999).
- [10] Houlberg W A, Shaing K C, Hirshman S P and Zarnstorff M C, *Bootstrap current and neoclassical transport in tokamaks of arbitrary collisionality and aspect ratio*, Phys. Plasmas **4** (1997)